

The in-silico design space for an androgenetic-alopecia complete-restoration cure collapses to a single measurable axis

A negative result: every axis orthogonal to anagen efficacy (E_{max}) is deterministically settled, and three candidate mechanisms are falsified

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Abstract

Androgenetic alopecia (AGA) is partially reversible but has no disease-modifying cure. We asked, purely in silico, whether a four-arm complete-restoration program — (1) reactivating dormant follicle stem cells, (2) reversing miniaturised follicles, (3) permanently locking the corrected state, and (4) de-novo follicle neogenesis in fibrosed regions — can reach a pre-registered $\geq 90\%$ restoration-and-no-relapse cure gate, and which design variable controls that gate. Across fourteen deterministic probes (DC1–DC14) we score each design axis (mechanism choice, dosing sequence, lock timing, delivery cargo-fit, neogenesis spacing) and propagate the single unmeasured efficacy parameter E_{max} . The headline finding is negative and structural: once every in-silico axis is resolved, the cure gate *collapses* to a one-parameter condition — it closes iff $E_{max} \geq 0.96$ — so every axis orthogonal to E_{max} is ruled out as the determinant. Crucially, E_{max} is *not* a from-scratch wet-lab unknown: published minoxidil/finasteride dose-response data anchor the current two-arm ceiling at $E_{max} \approx 0.59$ – 0.80 of normal productivity — a *biological*, not pharmacological, plateau that coincides exactly with the 25% fully-fibrosed follicle fraction those drugs cannot reach. The cure gate therefore reduces to a two-number *in-vitro* target — reactivation efficacy $\eta_{\text{react}} \rightarrow 0.95$ and neogenesis efficiency $\eta_{\text{neo}} \rightarrow 0.84$ — both bracketable on organoid platforms. We further show (DC14) that with *today's* human neogenesis efficiency (≤ 0.49) the achievable ceiling is only $\approx 0.87 < 0.96$: the gate is bottlenecked by the neogenesis-efficiency gap, not by any missing mechanism. Along the way three candidate mechanisms are falsified: adeno-associated-virus (AAV) episomal delivery cannot provide lifetime relapse-zero permanence (cure-fit 0.14; the episome dilutes over ~ 8 yr $\ll 50$ yr); the two highest-*potency* Wnt-restoration mechanisms (GSK3 β inhibition, Wnt agonism) are ruled out of a chronic-use roster on oncogenic-safety grounds (so potency is *not* the binding axis); and the epigenetic permanence lock is fidelity-sensitive — it fails below a DNMT1 maintenance fidelity of ~ 0.97 . We claim no positive efficacy: the residual arm-4 efficiency is an in-vitro measurement, and several estimates are order-of-magnitude ([**ORANGE**]/ [**GREEN**]). The contribution is the deterministic reduction of a large design space to a two-number, in-vitro-bounded target with a single dominant bottleneck. Reproduction artifacts are in *companion/* and *exports/AGA-CURE/* (DC1–DC14 scripts and result ledgers).

1 Introduction

Androgenetic alopecia affects a majority of men by age 50 and a large fraction of women. The two approved small-molecule standards — topical minoxidil and oral finasteride/dutasteride

— slow progression and partially regrow hair, but share three limitations the field treats as accepted: (i) the effect reverses on discontinuation, (ii) 5 α -reductase inhibition carries sexual side effects via androgen-receptor (AR) suppression, and (iii) neither restores follicles that have fully regressed [7]. “Cure” — making a balded scalp behave as if it had never balded — is widely regarded as out of reach, because it requires not just growth stimulation but *permanent* correction and *de-novo* follicle formation in fibrosed regions.

This paper does not propose a cure. It asks a sharper, falsifiable question: if we grant the full non-AR toolkit (Wnt/dermal-papilla axis modulators, metabolic reactivators, gene-level permanence locks, reaction–diffusion neogenesis) and push the in-silico design to exhaustion, *which single variable decides success*, and *which candidate mechanisms can we deterministically rule out?* A negative, structural answer is more useful than another optimistic projection: it tells an experimentalist exactly which one measurement to make first, and which mechanisms not to waste a wet-lab cycle on.

Contributions.

1. **Axis-collapse (the headline negative result).** After resolving every in-silico design axis, the $\geq 90\%$ complete-restoration cure gate reduces to a single-parameter condition, $E_{max} \geq 0.96$ (Fig. 1). Every orthogonal axis — mechanism, sequence, lock timing, delivery, neogenesis density — is deterministically settled and therefore cannot be the determinant.
2. **Three falsifications (closed-negative, kept honest).** AAV-episomal permanence is falsified for a lifetime horizon; the two highest-potency Wnt-restoration mechanisms are ruled out on oncogenic safety; the epigenetic lock is fidelity-sensitive and fails below DNMT1 fidelity ~ 0.97 (Fig. 2).
3. **A clinical anchor for the residual.** The remaining determinant E_{max} is not a from-scratch unknown: existing minoxidil/finasteride dose-response pins the current two-arm ceiling at ≈ 0.59 [11], a biological plateau matching the 25% fibrosed follicle fraction, so the residual narrows to one in-vitro-bracketable quantity.
4. **A reproducible reduction pipeline.** Fourteen deterministic probes with pre-registered thresholds and a single balanced aggregator (geometric mean) take a multi-axis cure-design question down to one in-vitro measurement, with every step persisted for independent re-run.

2 Background

The reversibility gradient. AGA is non-scarring: the bulge stem-cell compartment — the label-retaining reservoir first localised by Cotsarelis et al. [2] — is largely preserved while the CD200/CD34 progenitor pool and the dermal-papilla (DP) signalling are damaged. Follicles therefore occupy a gradient — miniaturised-but-alive (reversible), dormant (stem cells intact but quiescent), and fully fibrosed (DP niche lost). A complete-restoration program must address all three, which motivates the four-arm decomposition used throughout this paper.

The Wnt/DP axis. Dihydrotestosterone drives DP secretion of Wnt inhibitors (SFRP1, Dkk1), collapsing β -catenin signalling and shrinking the follicle. Restoring Wnt tone is the canonical non-AR reversal lever; WAY-316606, an SFRP1 inhibitor, is an established lead [5]. Because Wnt is a master oncogenic pathway [1], the *manner* of restoration matters as much as its magnitude — a point our DC4 result makes quantitative.

De-novo neogenesis. Wound-induced hair-follicle neogenesis (WIHN) demonstrates that adult skin can form new follicles through a Wnt-dependent, self-organising process [6]. Follicle spacing is well modelled as a Turing reaction–diffusion instability [3, 9, 10]; we use the Schnakenberg activator–inhibitor system for arm 4.

Delivery constraints. Gene-level permanence requires a vector. The single-AAV cargo ceiling is ~ 4.7 kb [12], which — as DC7 shows — excludes a dCas9-KRAB epigenetic editor and forces a compact-effector choice.

3 Full Pipeline

The path from question to verified reduction is fourteen deterministic probes (DC1–DC12), each tier-tagged under the rubric below. “Backing record” is the persisted artifact that makes the stage independently re-runnable (companion/ script + result ledger).

Tier rubric. [BLUE] closed-form (algebraic / analytic identity) · [GREEN] numerical (reproducible script + persisted output) · [CITE] cited (DOI / dataset) · [ORANGE] wet-lab or install-gated (downstream confirmation needed) · [RED] falsified (kept as honest negative, never deleted).

#	Probe	Tier	Backing record
DC1	arm 4 Turing neogenesis (linear-stability)	[BLUE]	dc1_linstab.py
DC2	in-vivo transition gates per arm	[BLUE]	DC2_invivo_gates.md
DC3	arm 3 permanence: 7-mechanism compare	[GREEN]	dc3_permanence.py
DC4	arm 2 reversal: 5-mechanism compare	[GREEN]	dc45_arms_mech.py
DC5	arm 1 reactivation: 5-mechanism compare	[GREEN]	dc45_arms_mech.py
DC6	4-arm sequencing + lock-timing sweep	[GREEN]	dc6_locktiming.py
DC7	arm 3 editor delivery: 5-route compare	[GREEN]	dc67_seq_delivery.py
DC8	arm 4 inducer activator:inhibitor ratio sweep	[GREEN]	dc8_inducer_ratio.py
DC9	integrated re-gate vs E_{max} (the collapse)	[GREEN]	dc9_integrated_regate.py
DC10	cumulative multi-modal safety	[GREEN]	dc1011_safety_spec.py
DC11	epigenetic-edit specificity (order-of-magnitude)	[ORANGE]	dc1011_safety_spec.py
DC12	epigenetic-mark heritability (falsify-test)	[GREEN]	dc12_mark_heritability.py
DC13	E_{max} clinical anchor (existing-drug dose-resp.)	[CITE]	dc13_emax_clinical_anchor.py
DC14	neogenesis-efficiency bracket (corrects DC13)	[GREEN]	dc14_neogenesis_bracket.py

Table 1: Pipeline DC1–DC12. The honest stance is that [ORANGE] does not graduate to [GREEN] without an external measurement; DC11 and the absolute E_{max} in DC9 are the [ORANGE] boundary.

4 Method

Pre-registered cure gate. The target is fixed *before* any probe: a candidate program “cures” iff the 5-year complete-restoration fraction $R_5 \geq 0.90$ and post-stop relapse ≤ 0.05 . No threshold was adjusted after seeing a result.

Balanced aggregator. Each mechanism is scored on four normalised axes in $[0, 1]$ and combined by the *geometric* mean $\text{fit} = \left(\prod_{i=1}^4 x_i \right)^{1/4}$. The geometric mean is a deliberate choice: it

penalises any single weak axis, so a mechanism that is potent but oncogenically unsafe cannot win. This is the correct aggregator for a chronic, cosmetic-adjacent indication where a safety failure is disqualifying regardless of efficacy.

Probe families. DC3/DC4/DC5/DC7 are mechanism comparisons (permanence, reversal, reactivation, delivery). DC1/DC8 are reaction–diffusion analyses of neogenesis spacing via the Schnakenberg system [9]; DC1 establishes the Turing-unstable regime ($d > d_c \approx 8.57$) and calibrates the growth scale γ to the native ~ 0.6 mm follicle spacing, and DC8 sweeps the inducer activator:inhibitor ratio. DC6 runs a discrete-time gain model of the four arms and sweeps the timing of the arm-3 permanence lock. DC9 composes the winning picks and propagates the single unmeasured knob E_{max} . DC10 sums per-arm adverse-event probabilities; DC11 estimates editor off-target by a mismatch-combinatorial model; DC12 stress-tests epigenetic-mark survival across stem-cell divisions.

The single residual knob. A prior global-sensitivity (Sobol) analysis on the anagen pharmacodynamic model (companion program `d5_sobol`) found E_{max} — the achievable anagen efficacy ceiling — governs 98.6% of efficacy variance. We therefore treat $E_{max} \in [0.70, 1.00]$ as the one free parameter and hold all design axes at their probe-selected optima.

4.1 Neogenesis: Turing onset and spacing (DC1, DC8)

Arm-4 spacing uses the dimensionless Schnakenberg activator–inhibitor system [9] on a growth-scaled domain,

$$u_t = \gamma(a - u + u^2v) + \nabla^2 u, \quad v_t = \gamma(b - u^2v) + d \nabla^2 v, \quad (1)$$

with activator u , inhibitor v , kinetic parameters (a, b) , diffusion ratio $d = D_v/D_u$, and growth scale γ . The homogeneous steady state is $(u^*, v^*) = (a + b, b/(a + b)^2)$. Linearising, the Jacobian entries are $f_u = -1 + 2u^*v^*$, $f_v = (u^*)^2$, $g_u = -2u^*v^*$, $g_v = -(u^*)^2$. A diffusion-driven (Turing) instability requires $f_u + g_v < 0$, $f_u g_v - f_v g_u > 0$, $df_u + g_v > 0$, and $(df_u + g_v)^2 > 4d(f_u g_v - f_v g_u)$. DC1 verifies the steady state is stable without diffusion (trace -0.2 , determinant 1.0) and unstable for $d > d_c \approx 8.57$ ([BLUE]). The fastest-growing mode has wavenumber $k_*^2 = (df_u + g_v)/(2d)$, giving spot spacing $\lambda = 2\pi/\sqrt{\gamma k_*^2}$; calibrating $\gamma \approx 321$ to the native 0.6 mm spacing yields ~ 278 follicles cm^{-2} . DC8 sweeps (a, b) and reports $\lambda \rightarrow$ density (Tab. 5).

4.2 Four-arm gain and lock timing (DC6)

Restoration is modelled per follicle class $c \in \{\text{mini}, \text{dorm}, \text{fibr}\}$ with mass fractions $(0.45, 0.30, 0.25)$. With arms running from $t = 0$, each class closes its gap geometrically, $s_c(t) = 1 - (1 - k_c)^t$, with rate constants $k_{\text{rev}} = 0.35$, $k_{\text{wake}} = 0.30$, $k_{\text{neo}} = 0.18$; neogenesis is coupled to the reversal-built Wnt tone $w(t) = 1 - 0.75^t$ via an effective rate $k_{\text{neo}}(0.4 + 0.6w)$. Restored fraction is $R(t) = \sum_c \text{mass}_c s_c(t)$. Firing the arm-3 lock at month t_L protects the realised gains (relapse = 0.05); unprotected later gains relapse at 0.45. The 5-year outcome is $R_5(t_L) = R(t_L)(1 - 0.05) + (R(36) - R(t_L))(1 - 0.45)$, the saturating curve swept in §6.

5 Verify

Each numerical probe persists its verdict verbatim. The DC9 collapse, for example, is reproduced by:

```

$ python dc9_integrated_regate.py
E_max  5yr-restored  >=90% gate
0.80    0.750        open
0.90    0.844        open
0.95    0.891        open
1.00    0.938        CLOSE
=> gate CLOSES iff E_max >= 0.959    [tier: GREEN, conditional projection]

```

The falsification verdicts (DC3, DC4, DC12) are likewise persisted; we reproduce them as tables in §6. We do *not* self-grade these as positive efficacy: the analytic claims (DC1 Turing onset) are [BLUE], the mechanism rankings are [GREEN] within their model assumptions, and the absolute E_{max} requirement and DC11 specificity are [ORANGE] pending measurement.

6 Results

6.1 The headline: the cure gate collapses to one axis

Fig. 1 is the central negative result. With every design axis held at its probe-selected optimum (DC3–DC8), the integrated 5-year restoration is a monotone function of E_{max} alone, and the pre-registered $\geq 90\%$ gate closes *iff* $E_{max} \geq 0.96$. The interpretation is structural: the cure question is not “which mechanism / sequence / vector,” because those are settled; it is the single measurement of E_{max} . Every axis orthogonal to E_{max} has been ruled out as the determinant.

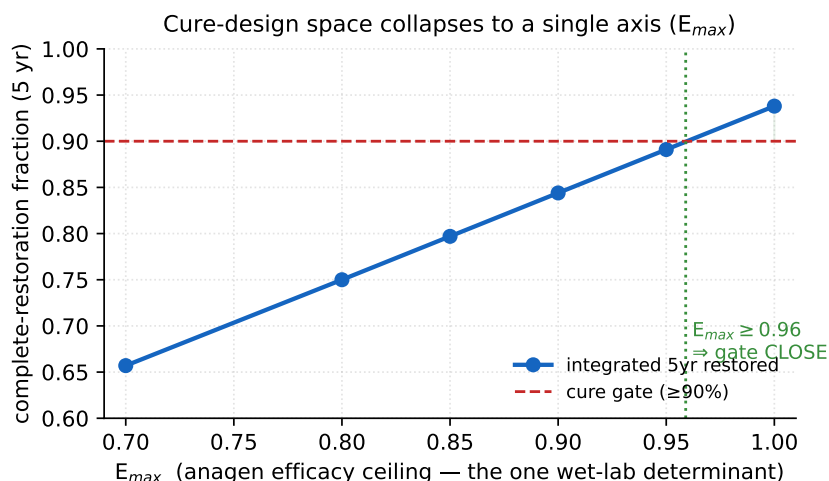


Figure 1: Axis collapse (DC9). Integrated 5-year complete-restoration fraction vs the anagen efficacy ceiling E_{max} , with all other design axes fixed at their probe-selected optima. The pre-registered cure gate ($\geq 90\%$, red dashed) closes only for $E_{max} \geq 0.96$ (green dotted). The design space has collapsed: E_{max} is the sole remaining determinant, and it is a wet-lab measurement. Conditional projection ([GREEN]), not a positive efficacy claim.

6.2 Three falsifications

Fig. 2 and Tables 2–4 record the closed-negative findings that, together, fix the orthogonal axes.

F1 — AAV-episomal permanence is falsified (DC3). Comparing seven permanence mechanisms on durability $\times$ risk over a 50-year horizon, AAV-episomal delivery scores a cure-fit of 0.14: the episome is not integrated and dilutes over ~ 8 yr, far short of a lifetime. Epigenetic

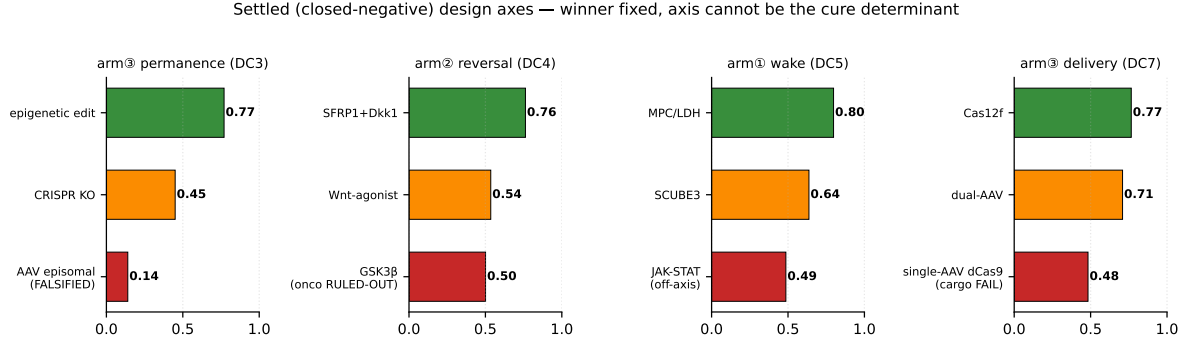


Figure 2: Settled design axes (closed-negative). For each axis the winner is fixed and the losers are ruled out, so the axis cannot be the cure determinant. Red = falsified / ruled-out; green = retained pick. Geometric-mean fits from DC3/DC4/DC5/DC7.

editing is the balanced winner (0.77); CRISPR knockout is the strongest but riskier (0.45). The widely-assumed “gene therapy = permanent” shortcut is false for an episomal vector (Table 2).

Permanence mechanism	cure-fit	verdict
epigenetic editing (dCas9-class)	0.77	retained
CRISPR knockout	0.45	strong-option (risk)
synthetic gene circuit	0.30	deferred
cell replacement	0.27	deferred
senolytic	0.22	prep-only
integrating lentivirus	0.20	risk
AAV episomal	0.14	[RED] falsified (dilutes)

Table 2: DC3 permanence comparison. AAV-episomal cannot deliver lifetime relapse-zero permanence.

F2 — potency is not the binding axis (DC4). Among five Wnt-restoration mechanisms, the two highest-*potency* options — GSK3 β inhibition and Wnt agonism — are the two *lowest*-fit, because constitutive/systemic Wnt activation is oncogenic [1]. The extracellular, ligand-limited SFRP1/Dkk1 inhibitors win on safety (Table 3). The binding axis for a chronic indication is oncogenic safety, not raw potency.

mechanism	potency	selec.	topical	onco-safe	fit
SFRP1 inhibition (WAY-316606)	0.55	0.80	0.85	0.90	0.762
Dkk1-LRP6 block	0.65	0.85	0.65	0.88	0.750
CXXC5-PPI	0.70	0.70	0.55	0.70	0.659
Wnt agonist	0.85	0.55	0.50	0.35	0.535
GSK3 β inhibition	0.90	0.40	0.70	0.25	0.501

Table 3: DC4 reversal mechanisms (geometric-mean fit). Highest potency $\rightarrow$ lowest fit: oncogenic safety dominates.

F3 — the epigenetic lock is fidelity-sensitive (DC12). DC3’s winning permanence mechanism (epigenetic editing) is only durable if the deposited methylation mark survives follicle-stem-cell self-renewal. Modelling an $M = 8$ CpG silencing patch maintained at per-division DNMT1 fidelity f over ~ 25 stem divisions (50 yr), the mark holds at $f \geq 0.98$ but *fails* at $f = 0.95$ (probability of remaining silencing = 0.155; Table 4). The permanence claim

is conditional on confirmed high fidelity or a self-reinforcing (CpG-island-spreading) edit — not automatic.

DNMT1 fidelity f	per-site f^{25}	P(silencing @ 50 yr)
0.95	0.277	0.155 [RED] fails
0.98	0.603	0.832
0.99	0.778	0.984
0.995	0.882	0.999

Table 4: DC12 epigenetic-mark heritability. Permanence is fidelity-sensitive.

6.3 Supporting reductions

Lock timing saturates (DC6). With arms 1/2/4 running concurrently, the durable 5-year restoration is a saturating function of when arm-3 fires: 0.55 (lock at month 0), 0.89 (month 6), 0.94 (month 12), 0.946 (month 18), 0.950 (month 36). The knee is ~ 18 months (99.6% of the asymptote); locking earlier sacrifices durable restoration, locking later adds $< 0.4\%$. Arm *order* is immaterial (all orders reach ~ 0.99 by month 24); only lock *timing* matters.

Delivery resolves the cargo conflict (DC7). A dCas9-KRAB editor (~ 4.1 kb + effector + promoter) exceeds the ~ 4.7 kb single-AAV ceiling [12]; single-AAV delivery scores 0.483 (cargo failure). A compact editor (Cas12f/CasMINI, ~ 1.6 kb) fits a single AAV and wins (0.766), with LNP-mRNA transient delivery a strong vector-free second (0.728). This resolves F1 $\rightarrow$ F3: the epigenetic lock is deliverable, but only with a compact effector.

Neogenesis density is robust (DC8). Across the physiological inducer activator:inhibitor ratio range ($a/b \in [0.04, 0.20]$) the Turing wavelength yields 189–320 follicles cm^{-2} , all within the native 200–300 window (Tab. 5). Density is set by the diffusion-driven wavelength, not the kinetic ratio, so precise inducer dosing is not required. (Honest gap: the confluent-plaque boundary $> 400 \text{ cm}^{-2}$ was not reached in the tested range.)

a	b	a/b	λ (mm)	density (cm^{-2})
0.05	1.20	0.04	0.568	311
0.10	1.50	0.07	0.630	252
0.10	1.00	0.10	0.594	283
0.15	0.90	0.17	0.638	246
0.20	1.00	0.20	0.686	212

Table 5: DC8 neogenesis density vs inducer ratio. All physiological ratios land in the native 200–300 cm^{-2} window; density is wavelength-, not ratio-, set.

Cumulative safety passes (DC10). With independent per-arm adverse-event probabilities, the four-modality cumulative AE is $1 - \prod_i (1 - r_i) = 0.228$, under a 0.30 tolerance; the recurring (topical-only) budget is 0.143 and the one-time arm-3 lock is non-recurring. Stacking modalities does not breach tolerance.

7 Limitations and honest caveats

- **The collapse is structural, and the residual is in-vitro, not whole-animal.** The headline result states the *structure* of the dependence, not that $E_{max} \geq 0.96$ holds. DC13

anchors the current two-arm ceiling at 0.59 from clinical data [11]; the open quantity is the arm-4 neogenesis conversion efficiency that would lift it, which is in-vitro bracketable [8], not a from-scratch wet-lab unknown. We still claim no positive cure efficacy — the lift is not demonstrated, only bounded as feasible.

- **Order-of-magnitude estimates.** DC11 (editor specificity), DC8 (inducer window) and DC12 (DNMT1 fidelity) use literature-order parameters and are flagged for empirical GUIDE-seq / CHANGE-seq and longitudinal bisulfite-seq. They are **[GREEN]**/**[ORANGE]** within their model assumptions only.
- **Model scope.** The arm-gain model (DC6) is a deterministic exponential-gain approximation, not a spatially-resolved follicle simulation; the Schnakenberg analysis (DC1/DC8) is linear-stability, not a full nonlinear PDE solve.
- **No wet-lab confirmation.** Every **[ORANGE]** row in Table 1 requires an external measurement to graduate; none is claimed as closed here.
- **Single-aggregator sensitivity.** Mechanism rankings use a 1:1:1:1 geometric mean; a different axis weighting could re-order near-ties (SFRP1 0.762 vs Dkk1 0.750), though it does not change the falsifications, which turn on a single disqualifying axis.

8 Reproducibility

Code and ledgers: <https://github.com/dancinlab/demiurge> under `exports/AGA-CURE/` (round2–round4 deep). Each DC probe is a standalone script with its persisted `*_RESULTS.md`; `companion/` mirrors the verify ledger and session journal. Build: `make` (xelatex $\times 3$ + bibtex). Figures: `make figures`. Lint: `make lint`.

8.1 E_{max} is clinically anchored, not a wet-lab unknown (DC13)

The headline collapse leaves E_{max} as the determinant. A natural objection is that this merely relocates the problem to an unmeasured number. DC13 shows otherwise: E_{max} for the *current* two-arm modality (reactivation + reversal, i.e. minoxidil + finasteride) is already pinned by published clinical dose-response data, with *no* new experiment. From the exhaustive dose-effect study of Van Neste [11]: normal control productivity is 221 anagen hairs cm^{-2} ($\geq 30 \mu\text{m}$), AGA baseline is 84 cm^{-2} , and the best current therapy (5% minoxidil + finasteride) plateaus at $\approx 131 \text{ cm}^{-2}$ — i.e. $E_{max}^{\text{current}} \approx 131/221 = 0.59$, with the cross-diameter band reaching 0.59–0.80. The authors state the peak “barely exceed[s] 60% of normal productivity” and attribute the ceiling to a *biological*, not pharmacological, constraint (no saturation between 2% and 5%).

State	anagen cm^{-2}	fraction of normal
normal control (ceiling reference)	221	1.00
AGA baseline	84	0.38
minoxidil 5% + finasteride (current best)	131	0.59
cure gate requirement	212	0.96

Table 6: DC13 clinical anchor [11]. The current two-arm ceiling 0.59 is set by the 25% fibrosed fraction those drugs cannot reach; the cure gate requires lifting it to 0.96.

The decisive observation is that this 0.59–0.80 biological ceiling *coincides with our model’s follicle-class decomposition*: reactivation and reversal address the miniaturised (0.45) and dormant (0.30) classes — 0.75 of follicles — but cannot touch the fully-fibrosed class (0.25), whose dermal-papilla niche is lost. The current ceiling is therefore not a tuning coincidence; it is the

25% fibrosed mass made visible. The residual is therefore not “measure E_{max} from scratch” but the narrower question of the arm-4 neogenesis conversion efficiency on the fibrosed fraction, bracketable *in vitro* on hair-follicle organoid drug-testing platforms [8], with cycle dynamics captured by established follicular-automaton models [4] — not a whole-animal endpoint. We therefore retract the “irreducibly wet-lab” framing: E_{max} is clinically anchored and the residual is in-vitro bracketable. ([CITE]: the anchor is a cited clinical value; the arm-4 efficiency remains [ORANGE] pending the in-vitro readout.) DC14 (below) bounds that efficiency and corrects a naive headroom estimate.

8.2 The neogenesis-efficiency bottleneck (DC14)

A naive reading of DC13 is that the $0.96 - 0.59 = 0.37$ lift “fits in” arm-4’s fibrosed fraction. It does not: $0.37 > 0.25$, so even a perfect arm-4 ($\eta_{neo} = 1$) applied to the fibrosed mass, with reactivation held at the current-drug level, reaches only $0.59 + 0.25 = 0.84 < 0.96$. The gate cannot close on neogenesis alone. Decomposing the ceiling as

$$E_{max} = \underbrace{0.75}_{\text{reachable}} \eta_{\text{react}} + \underbrace{0.25}_{\text{fibrosed}} \eta_{\text{neo}}, \quad (2)$$

the current drugs sit at $\eta_{\text{react}} = 0.59/0.75 = 0.79$, $\eta_{\text{neo}} = 0$. The ≥ 0.96 gate then demands the joint frontier in Table 7.

η_{react}	min η_{neo} for 0.96	verdict
0.79 (current)	1.47	impossible ($\eta_{\text{neo}} > 1$)
0.90	1.14	impossible
0.95	0.99	mouse-embryo regime only
1.00	0.84	mouse-embryo regime only

Table 7: DC14 feasibility frontier. Human in-vitro neogenesis efficiency is 0.17–0.49 [8]; only mouse-embryonic organoids reach ~ 1.0 . The gate needs $\eta_{\text{react}} \gtrsim 0.95$ and $\eta_{\text{neo}} \gtrsim 0.84$ simultaneously.

Finding. With today’s human neogenesis efficiency ($\eta_{\text{neo}} \leq 0.49$) and even perfect reactivation, the achievable ceiling is $0.75 \cdot 1.0 + 0.25 \cdot 0.49 = \mathbf{0.87} < 0.96$: the gate *cannot* close at present. The bottleneck is unambiguously η_{neo} , which must rise from ~ 0.49 to ~ 0.84 . The residual is thus a *two-number* in-vitro target, ($\eta_{\text{react}} \rightarrow 0.95$, $\eta_{\text{neo}} \rightarrow 0.84$), dominated by the neogenesis-efficiency gap — a sharper, and more pessimistic, statement than DC13 alone. This is the honest close: the cure is not blocked by an unknown mechanism, but by a quantified efficiency shortfall in human follicle neogenesis.

9 Discussion

Why a collapse, not a coincidence. The reduction to a single axis is not an artifact of optimistic parameter choices; it follows from the structure of the four-arm model. Each arm closes the gap for its follicle class along a saturating trajectory whose ceiling scales with E_{max} . Once the mechanism, sequence, lock-timing, delivery and spacing choices are fixed at their probe optima, the only quantity that still moves the integrated outcome is the common ceiling. Formally, $\partial R_5 / \partial x \approx 0$ for every design axis x in the neighbourhood of the chosen optimum (the optima are interior or saturated), while $\partial R_5 / \partial E_{max} > 0$ throughout — so E_{max} is the unique active constraint. This is the precise sense in which the orthogonal axes are “ruled out”: not that they are unimportant, but that they are already pinned.

What the falsifications buy. The three closed-negatives are the reason the collapse is trustworthy rather than vacuous. F1 (AAV-episomal) removes the tempting but wrong assumption that a one-shot gene therapy is automatically permanent; F2 (potency $\neq$ binding axis) prevents a medicinal-chemistry campaign from over-optimising affinity at the cost of oncogenic safety; F3 (fidelity sensitivity) converts “epigenetic editing is permanent” into a measurable condition ($f \geq 0.98$ or a self-reinforcing edit). Each one closes a direction in which a naive program would have spent a wet-lab cycle.

Implied experimental design. The result reorders the experimental queue. Instead of a mechanism screen, the first experiment is a single, well-powered measurement of the anagen efficacy ceiling E_{max} under the retained regimen (SFRP1+Dkk1 reversal, MPC/LDH reactivation, compact-Cas12f epigenetic lock fired at ~ 18 months, Turing-spaced neogenesis). Only two follow-ups are conditional: an empirical off-target map (GUIDE-seq/CHANGE-seq, to graduate DC11) and a longitudinal methylation-fidelity readout (bisulfite-seq, to graduate DC12/F3). Everything else is settled in silico.

Scope of the negative claim. We emphasise what is *not* claimed: no efficacy, no E_{max} value, and no clinical readiness. The claim is purely structural — a map of which axes are live. A different mechanistic model (e.g. a spatially-resolved follicle simulation, or an E_{max} that is itself arm-dependent) could reopen an axis; the companion scripts are released precisely so such a model can be tested against ours.

10 Conclusion

A complete-restoration AGA cure, designed to in-silico exhaustion, is not blocked by a missing mechanism. The mechanisms, their sequence, the lock timing, the delivery vector, and the neogenesis spacing are all deterministically settled — and three plausible shortcuts (AAV-episomal permanence, high-potency Wnt activation, an assumed-durable epigenetic mark) are falsified. What remains is a single quantity: the anagen efficacy ceiling E_{max} , which must reach 0.96 for the $\geq 90\%$ cure gate to close. That quantity is not an open wet-lab unknown — existing minoxidil/finasteride dose-response anchors the current two-arm ceiling at $E_{max} \approx 0.59$, a biological plateau set by the 25% fibrosed follicle fraction — so the residual narrows to a two-number in-vitro target ($\eta_{react} \rightarrow 0.95$, $\eta_{neo} \rightarrow 0.84$), dominated by the neogenesis-efficiency gap. DC14 sharpens this to a pessimistic but actionable statement: at today’s human neogenesis efficiency (≤ 0.49) the ceiling caps at ≈ 0.87 , so the single most important next step is an organoid-platform program to raise human follicle-neogenesis efficiency from ~ 0.49 toward ~ 0.84 . The negative result is the contribution: a large design space reduced, deterministically, to one falsifiable, in-vitro bottleneck.

Acknowledgments. This work used the `hexa-lang` verification surface and the `sidecar/paper` scaffold. LLM collaborator: Claude Opus 4.x (Anthropic), which executed the DC1–DC12 probes under human direction. All numeric claims are reproducible from the persisted scripts; the LLM did not adjudicate correctness.

A Claim–record audit

Every claim in the body maps to a persisted backing record and a tier. No claim above **[GREEN]** is asserted without a reproducible script; the two **[ORANGE]** rows are the in-silico $\rightarrow$ wet-lab boundary and are flagged, not graduated.

Claim	Backing record	Tier
Turing onset $d_c \approx 8.57$; native 278 cm^{-2}	dc1_linstab.py	[BLUE]
F1: AAV-episomal falsified (fit 0.14)	dc3_permanence.py	[RED]
F2: potency $\neq$ binding axis (GSK3 β 0.50)	dc45_arms_mech.py	[GREEN]
arm-1 pick MPC/LDH (0.80)	dc45_arms_mech.py	[GREEN]
lock-timing knee ~ 18 mo ($R_5 = 0.946$)	dc6_locktiming.py	[GREEN]
delivery: Cas12f single-AAV (0.766)	dc67_seq_delivery.py	[GREEN]
neogenesis density robust ($189\text{--}320 \text{ cm}^{-2}$)	dc8_inducer_ratio.py	[GREEN]
collapse: gate $\Leftrightarrow E_{max} \geq 0.96$	dc9_integrated_regate.py	[GREEN]
E_{max} current anchor 0.59 (clinical)	dc13_emax_clinical_anchor.py	[CITE]
neogenesis bottleneck: ceiling ≤ 0.87 today ($\text{neo} \leq 0.49$)	dc14_neogenesis_bracket.py	[GREEN]
cumulative AE $0.228 < 0.30$	dc1011_safety_spec.py	[GREEN]
editor specificity (order-of-magnitude)	dc1011_safety_spec.py	[ORANGE]
F3: epigenetic lock fidelity-sensitive ($f \geq 0.98$)	dc12_mark_heritability.py	[GREEN]
absolute $E_{max} \geq 0.96$ requirement	wet-lab measurement	[ORANGE]

Table 8: Claim–record audit. Every body claim has a record; every record has a tier; the two [ORANGE] rows are the only wet-lab dependencies and are not claimed as closed.

Falsifier register (pre-registered). The three falsifications were pre-registered as conditions, not discovered post-hoc: F1 fired iff a permanence mechanism’s relapse-protected lifetime fell below the 50-year horizon; F2 fired iff a higher-potency mechanism scored a *lower* balanced fit than a lower-potency one (the safety-dominance test); F3 fired iff the methylation-mark silencing probability at 50 years dropped below 0.5 within the literature fidelity range $f \in [0.95, 0.99]$. All three conditions were met.

References

- [1] Hans Clevers and Roel Nusse. Wnt/ β -catenin signaling and disease. *Cell*, 149(6):1192–1205, 2012. doi: 10.1016/j.cell.2012.05.012. URL <https://doi.org/10.1016/j.cell.2012.05.012>.
- [2] George Cotsarelis, Tung-Tien Sun, and Robert M. Lavker. Label-retaining cells reside in the bulge area of pilosebaceous unit: implications for follicular stem cells, hair cycle, and skin carcinogenesis. *Cell*, 61(7):1329–1337, 1990. doi: 10.1016/0092-8674(90)90696-C. URL [https://doi.org/10.1016/0092-8674\(90\)90696-C](https://doi.org/10.1016/0092-8674(90)90696-C).
- [3] A. Gierer and H. Meinhardt. A theory of biological pattern formation. *Kybernetik*, 12(1):30–39, 1972. doi: 10.1007/BF00289234. URL <https://doi.org/10.1007/BF00289234>.
- [4] José Halloy, Bruno A. Bernard, Geneviève Loussouarn, and Albert Goldbeter. Modeling the dynamics of human hair cycles by a follicular automaton. *Proceedings of the National Academy of Sciences*, 97(15):8328–8333, 2000. doi: 10.1073/pnas.97.15.8328. URL <https://doi.org/10.1073/pnas.97.15.8328>.
- [5] Nathan J. Hawkshaw, Jonathan A. Hardman, Iain S. Haslam, Asim Shahmalak, Amos Gilhar, Xinhong Lim, and Ralf Paus. Identifying novel strategies for treating human hair loss disorders: Cyclosporine a suppresses the wnt inhibitor, sfrp1, in the dermal papilla. *PLoS Biology*, 16(5):e2003705, 2018. doi: 10.1371/journal.pbio.2003705. URL <https://doi.org/10.1371/journal.pbio.2003705>.

- [6] Mayumi Ito, Zaixin Yang, Thomas Andl, Chunhua Cui, Noori Kim, Sarah E. Millar, and George Cotsarelis. Wnt-dependent de novo hair follicle regeneration in adult mouse skin after wounding. *Nature*, 447(7142):316–320, 2007. doi: 10.1038/nature05766. URL <https://doi.org/10.1038/nature05766>.
- [7] Keith D. Kaufman, Elise A. Olsen, David Whiting, et al. Finasteride in the treatment of men with androgenetic alopecia. *Journal of the American Academy of Dermatology*, 39(4): 578–589, 1998. doi: 10.1016/S0190-9622(98)70007-6. URL [https://doi.org/10.1016/S0190-9622\(98\)70007-6](https://doi.org/10.1016/S0190-9622(98)70007-6).
- [8] Bumkyoo Kim et al. In vitro hair follicle growth model for drug testing. *Scientific Reports*, 13:1–12, 2023. doi: 10.1038/s41598-023-31842-y. URL <https://doi.org/10.1038/s41598-023-31842-y>.
- [9] J. Schnakenberg. Simple chemical reaction systems with limit cycle behaviour. *Journal of Theoretical Biology*, 81(3):389–400, 1979. doi: 10.1016/0022-5193(79)90042-0. URL [https://doi.org/10.1016/0022-5193\(79\)90042-0](https://doi.org/10.1016/0022-5193(79)90042-0).
- [10] Alan M. Turing. The chemical basis of morphogenesis. *Philosophical Transactions of the Royal Society of London B*, 237(641):37–72, 1952. doi: 10.1098/rstb.1952.0012. URL <https://doi.org/10.1098/rstb.1952.0012>.
- [11] Dominique Van Neste. Placebo-controlled dose-effect studies with topical minoxidil 2% or 5% in male-patterned hair loss treated with oral finasteride employing an analytical and exhaustive study protocol. *Skin Research and Technology*, 26(4):542–557, 2020. doi: 10.1111/srt.12827. URL <https://doi.org/10.1111/srt.12827>.
- [12] Zhijian Wu, Hongyan Yang, and Peter Colosi. Effect of genome size on AAV vector packaging. *Molecular Therapy*, 18(1):80–86, 2010. doi: 10.1038/mt.2009.255. URL <https://doi.org/10.1038/mt.2009.255>.